WORKED SOLUTION

STEP II 2008, Question 6

Pure • Sum-to-Product Formulae

STEP II 2008, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q6

The key idea

A sum of a cosine and a sine with different arguments is awkward, but a sum of two cosines factorises into a product by the sum-to-product formula. Rewrite the sine as a cosine first; then the zeros of f are the zeros of two simple factors, and a point where *both* factors vanish is where the curve touches the axis. For the last part, remember that a cosine and a sine can each be at most 1.

Solution

Throughout, write

$$A = 2x + \frac{\pi}{3}, \qquad \sin\left(\frac{3x}{2} - \frac{\pi}{4}\right) = \cos\left(\frac{\pi}{2} - \frac{3x}{2} + \frac{\pi}{4}\right) = \cos B, \quad \text{where } B = \frac{3\pi}{4} - \frac{3x}{2},$$

so that $f(x) = \cos A + \cos B$.

Part (i)

The term $\cos(2x + \frac{\pi}{3})$ has period π, and $\sin(\frac{3x}{2} - \frac{\pi}{4})$ has period $\frac{2\pi}{3/2} = \frac{4\pi}{3}$. A common period must be a whole number of each: $p\pi = q \cdot \frac{4\pi}{3}$ means $3p = 4q$, and the smallest positive solution is $p = 4$, $q = 3$. So $f(x + 4\pi) = f(x)$ for all x, and the period of f is

$$4\pi.$$

Nothing smaller works: we shall see in part (iii) that $f(x) = 2$ exactly when $x = \frac{11\pi}{6} + 4t\pi$ for integer t, so the points where f reaches its maximum are spaced 4π apart, and any period must be a multiple of 4π.

Part (ii)

Using $\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$, with

$$\frac{A+B}{2} = \frac{1}{2}\left(\frac{x}{2} + \frac{13\pi}{12}\right) = \frac{x}{4} + \frac{13\pi}{24},$$
$$\frac{A-B}{2} = \frac{1}{2}\left(\frac{7x}{2} - \frac{5\pi}{12}\right) = \frac{7x}{4} - \frac{5\pi}{24},$$

we get

$$f(x) = 2\cos\left(\frac{x}{4} + \frac{13\pi}{24}\right)\cos\left(\frac{7x}{4} - \frac{5\pi}{24}\right).$$

So $f(x) = 0$ exactly when one of the two cosines is zero.

First factor. $\frac{x}{4} + \frac{13\pi}{24} = \frac{\pi}{2} + k\pi$ gives $x = -\frac{\pi}{6} + 4k\pi$. The only one in $[-\pi, \pi]$ is $x = -\frac{\pi}{6}$.

Second factor. $\frac{7x}{4} - \frac{5\pi}{24} = \frac{\pi}{2} + k\pi$ gives $\frac{7x}{4} = \frac{17\pi}{24} + k\pi$, so

$$x = \frac{(17 + 24k)\pi}{42}.$$

We need $-42 \le 17 + 24k \le 42$, so $k = -2, -1, 0, 1$, giving $x = -\frac{31\pi}{42}$, $-\frac{7\pi}{42} = -\frac{\pi}{6}$, $\frac{17\pi}{42}$, $\frac{41\pi}{42}$.

Combining, the zeros of f in $-\pi \le x \le \pi$ are

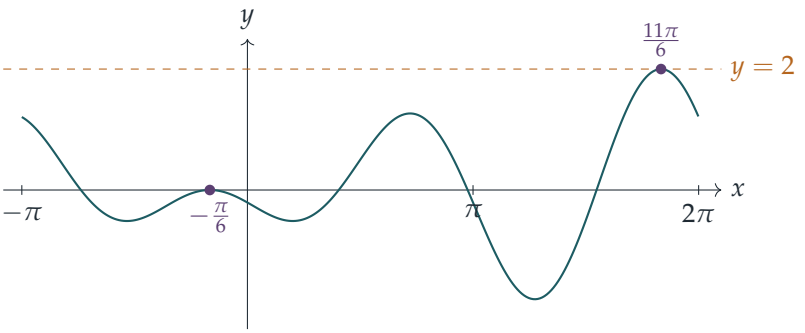
$$x = -\frac{31\pi}{42}, \quad -\frac{\pi}{6}, \quad \frac{17\pi}{42}, \quad \frac{41\pi}{42}.$$

Ask yourself Why should a value that appears in *both* lists be special?

At $x = -\frac{\pi}{6}$ both cosine factors are zero. Each is a cosine passing through a zero, so each changes sign there (its derivative is non-zero, since sin is ± 1 where cos is 0). Near $x = -\frac{\pi}{6}$, then, f behaves like a constant times $(x + \frac{\pi}{6})^2$: the product of two sign changes is no sign change. In fact, as x increases through $-\frac{\pi}{6}$, the first factor goes from positive to negative (its argument passes up through $\frac{\pi}{2}$) and the second from negative to positive (its argument passes up through $-\frac{\pi}{2}$), so f is negative on either side. Hence the curve touches the x-axis without crossing it at

$$x = -\frac{\pi}{6}.$$

At the other three zeros only one factor vanishes, so f changes sign and the curve crosses.



Part (iii)

Since $\cos A \leq 1$ and $\cos B \leq 1$, we have $f(x) \leq 2$, with equality if and only if $\cos A = 1$ *and* $\cos B = 1$. In terms of the original functions:

$$\cos\left(2x + \frac{\pi}{3}\right) = 1 \iff 2x + \frac{\pi}{3} = 2m\pi \iff x = -\frac{\pi}{6} + m\pi,$$
$$\sin\left(\frac{3x}{2} - \frac{\pi}{4}\right) = 1 \iff \frac{3x}{2} - \frac{\pi}{4} = \frac{\pi}{2} + 2n\pi \iff x = \frac{\pi}{2} + \frac{4n\pi}{3},$$

for integers m, n. Both hold when $-\frac{\pi}{6} + m\pi = \frac{\pi}{2} + \frac{4n\pi}{3}$; multiplying by $\frac{6}{\pi}$, this is $6m - 1 = 3 + 8n$, that is,

$$3m - 4n = 2.$$

One solution is $m = 2$, $n = 1$, and the general solution is $m = 2 + 4t$, $n = 1 + 3t$ (since 3 and 4 are coprime). So $f(x) = 2$ exactly when

$$x = -\frac{\pi}{6} + (2 + 4t)\pi = \frac{11\pi}{6} + 4t\pi, \qquad t \in \mathbb{Z}.$$

The only such value in $0 \leq x \leq 2\pi$ is

$$x = \frac{11\pi}{6}.$$

Check: $2 \cdot \frac{11\pi}{6} + \frac{\pi}{3} = 4\pi$ and $\frac{3}{2} \cdot \frac{11\pi}{6} - \frac{\pi}{4} = \frac{5\pi}{2}$, so $f = \cos 4\pi + \sin\frac{5\pi}{2} = 1 + 1 = 2$.

Checks

- The factorised form agrees with the original f at 10^4 random points (to 10^{-12}).
- Scanning $[-\pi, \pi]$ numerically finds sign changes only near $-\frac{31\pi}{42}$, $\frac{17\pi}{42}$ and $\frac{41\pi}{42}$; at $-\frac{\pi}{6}$ we find $f(-\frac{\pi}{6} \pm 10^{-3}) \approx -8.7 \times 10^{-7}$ on *both* sides, confirming a touching zero with no sign change.
- The maximum of f on $[0, 2\pi]$ is 2, attained at $x \approx 1.8333\pi = \frac{11\pi}{6}$.

Common slips

- Listing $-\frac{\pi}{6}$ twice, or missing that it comes from both factors, which is exactly the clue to the touching point.
- Solving $\cos A = 1$ and $\cos B = 1$ separately and stopping: both conditions must hold at the same x, which needs the integer equation $3m - 4n = 2$.

Beyond STEP

Part (iii) is really a question about when two periodic motions line up: the condition $3m - 4n = 2$ is a linear Diophantine equation, solvable because $\gcd(3, 4)$ divides 2. The same idea, applied to many frequencies at once, leads to Fourier series and to the question of when a sum of periodic functions is itself periodic (only when the periods are rationally related). For the number theory behind it, try Davenport, *The Higher Arithmetic*.

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